C 43190 **(Pages : 3)** **Name.....**

Reg. No.....

SECOND SEMESTER (CBCSS–UG) DEGREE EXAMINATION, APRIL 2023

Mathematics

MAT 2C 02—MATHEMATICS – 2

(2019 Admissions)

Time : Two Hours Maximum : 60 Marks

Section A

Answer any number of questions.
Each question carries 2 *marks.*
Maximum marks : 20.

1. Prove the product formula : $\cos\theta\cos\phi = \frac{1}{2}\left[\cos(\theta-\phi)+\cos(\theta+\phi)\right]$.
2. State the inverse function rule for finding the derivative of the inverse of a given function.
3. Write down the formula for area of surface of revolution.
4. State the comparison test for integrals.
5. Define absolutely convergent and conditionally convergent series.
6. Briefly explain Taylor's and Maclaurin's series.
7. Define linear independence.
8. Find the coordinates of the vector $u = \langle 3,-2,9\rangle$ relative to the orthonormal basis B for R^3 given in (1) of Example 1. Write u in terms of the basis B.
9. Write down the formula for cofactor expansion of an nxn matrix.
10. Describe Cramer's Rule briefly.
11. Define eigen value and eigen vector.
12. State Cayley Hamilton theorem.

(20 marks)

Turn over

Section B

Answer any number of questions.
Each question carries 5 *marks.*
Maximum marks : 30.

13. Prove the relation $\sinh^{-1} y = \ln\left(y + \sqrt{y^2+1}\right)$.

14. Find the length of the cardioid $r = 1 + \cos\theta (0 < \theta < 2\pi)$.

15. Find (a) $\lim_{n\to\infty}\left(\frac{n^2+1}{3n^2+n}\right)$

and (b) $\lim_{n\to\infty}\left(1 - \frac{3}{n} + \frac{n}{n+1}\right)$.

16. Show that $\lim_{x\to\infty} x^{1/x} = 1$.

17. Define a vector space.

18. Find the rank of

$$A = \begin{pmatrix} 1 & 1 & -1 & 3 \\ 2 & -2 & 6 & 8 \\ 3 & 5 & -7 & 8 \end{pmatrix}.$$

19. Show that the column vectors of the following matrix are orthogonal unit vectors :

$$A = \begin{pmatrix} \frac{1}{4} & -\frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{2}{3} \end{pmatrix}.$$

(30 marks)

Section C

Answer any **one** *question.*
Each question carries 10 *marks.*
Maximum 10 *marks.*

20. (a) Write down the Maclaurin series for $\sin x$.

    (b) Find the terms through cubic order in the Taylor series for $1/(1 + x^2)$ at $x_0 = 1$.

    (both in direct and indirect methods)

21. Diagonalize $A = \begin{pmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix}$.

(1 × 10 = 10 marks)